SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0223S

Revision 10

GE 1.5T 8 Channel
High Resolution Brain Array

GE Part Number: M3087JA

Invivo Part Number: 800223

Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF

Rev. 11/15/2000

Language Policy For Service Documentation (Dir. 2128126)

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- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

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- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇAO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE SIGNA 1.5T 8 Channel High Resolution Brain Array.

SHIPPING LIST – TABLE 1-1

Description	GE Part #	INVIVO #	Qty
1.5T 8 Ch HiRes Brain Array / Baseplate Assm.	2317112-2	101844	1
Phantom Positioner	2317112-3	102558	1
Operator Manual	2317112-4	500077	1
Service Manual	2317112-5	TR0223S	1
Thick Head Pad	2317112-9	102985	1
Thin Head Pad	2317112-10	102984	1
Spacer Pad	2317112-7	101622	1
Wedge Pad	2317115-8	101407	2

1-2 Compatibility

This coil is compatible with the following hardware configurations

- Signa Horizon and LX Horizon 1.5T with Excite Technology (8-channel)

1-3 Related Documentation

- Signa LX Service Methods CD, 2160623-1
- 8 Channel High Resolution Brain Array Operator's Manual, 2383610-4

1-4 Environmental Requirements

Storage Requirements

Coil should be stored flat in the scanner room.

Dimensions

Coil Dimension:	417.6 mm x 393.4 mm x 406.4 mm	(16.44 in x 15.49 in x 16.00 in)
Phantom Positioner:	225.6 mm x 228.6 mm x 23.9 mm	(8.88 in x 9.00 in x 0.94 in)

Weight

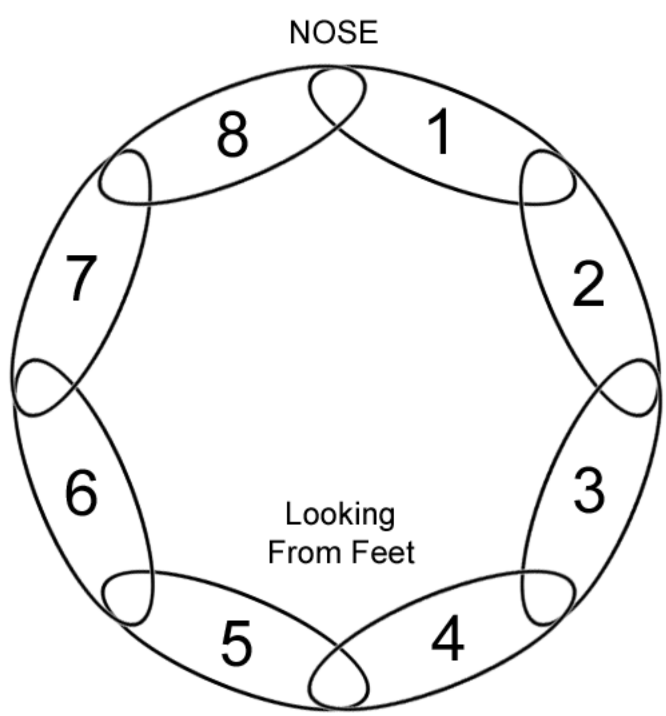
Coil Weight:	7.20 Kg	(16.0 lb)
Phantom Positioner:	0.45 Kg	(1.0 lb)

1-5 Theory of Operation

Refer to Figure 1 below and the Block Diagram on the following page.

The 1.5T 8 Channel High Resolution Brain Array consists of 8 elements. Each element is comprised of a loop arranged azimuthally around the head. Each loop is approximately 45° in azimuthal extent. The center of Loop #1 is located 22.5° rotated in the clockwise direction (as viewed from the feet) with respect to the nose. Each of the next seven loops is rotated clockwise by 45° successively. Loops #1 and #8 are located anteriorly while loops #4 and #5 are located posteriorly. Loops #2 and #3 are located on the patient Left side while Loops #6 and #7 are located on the patient Right side. During transmit, each loop is decoupled with both passive and active decoupling, with an additional active preamp protection network. During receive, signals from 1 – 8 are directed to separate ultra-low-input impedance preamp modules placed directly at the output of the loop, providing additional element decoupling.

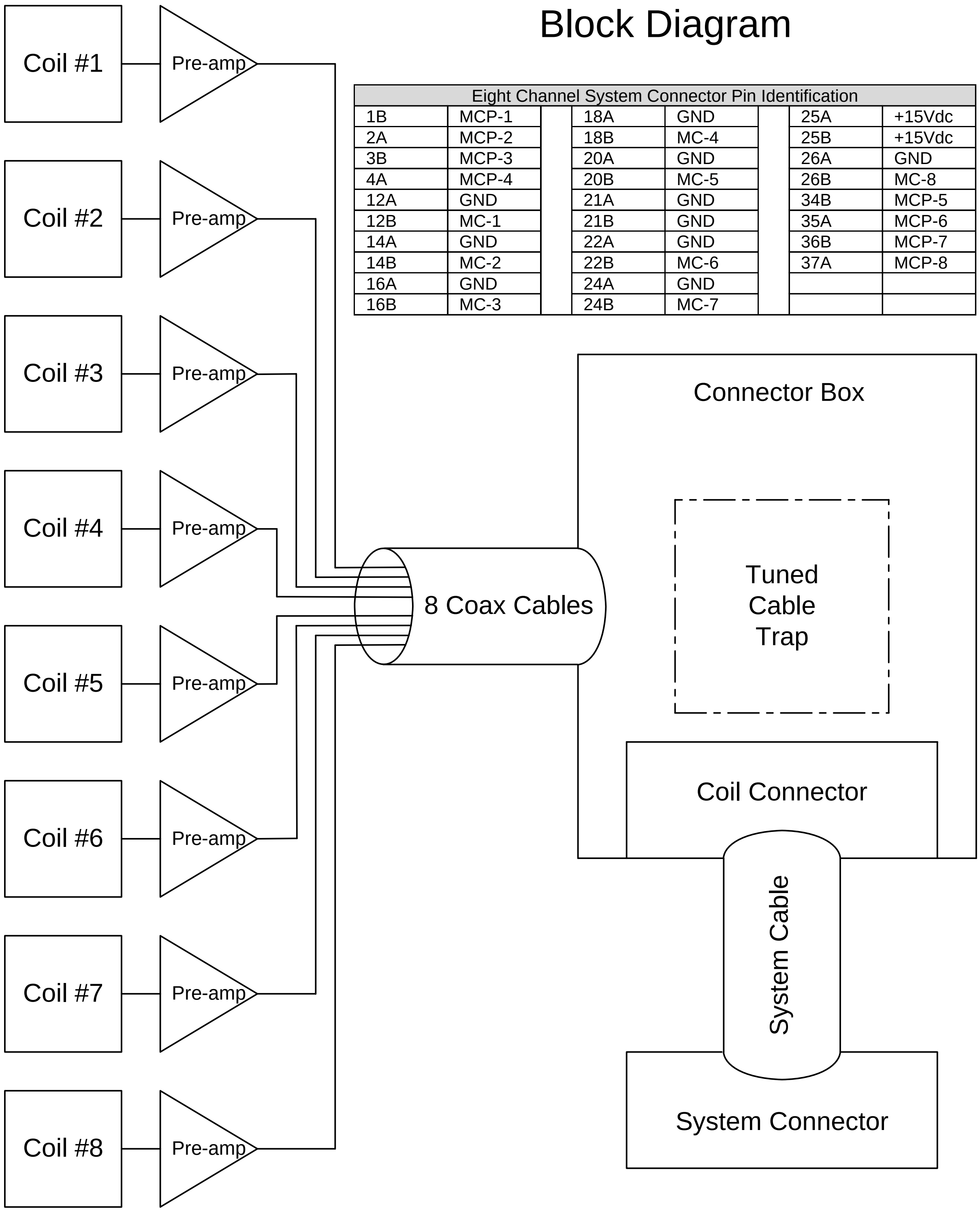
Figure 1



Loop Element Diagram

GE SIGNA 1.5T 8 Channel High Resolution Head Array Coil Block Diagram

Eight Channel System Connector Pin Identification					
1B	MCP-1	18A	GND	25A	+15Vdc
2A	MCP-2	18B	MC-4	25B	+15Vdc
3B	MCP-3	20A	GND	26A	GND
4A	MCP-4	20B	MC-5	26B	MC-8
12A	GND	21A	GND	34B	MCP-5
12B	MC-1	21B	GND	35A	MCP-6
14A	GND	22A	GND	36B	MCP-7
14B	MC-2	22B	MC-6	37A	MCP-8
16A	GND	24A	GND		
16B	MC-3	24B	MC-7		



SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None

2-1-2 Installing the Coil

The name for this coil is: **8HRBRAIN**

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

2. Perform Section 3 – Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.

2. Perform Section 3 – Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	INVIVO #	Qty
18cm Diameter Spherical TLT Phantom	46-265826G6	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2-2 QA Phantom and Coil Setup, and SNR Test Execution

1. Attach the base plate with 8 channel hi res brain coil to the cradle in the same position on the cradle as the standard head coil. Note the serial number of the coil.
2. Slide the coil back to permit access to the head holder as shown in Figure 3-2-2-1.
3. Place phantom ring on head holder as shown in Figure 3-2-2-2. The edge of the ring that goes toward the magnet has edges that overhang the head holder to secure the ring in position.
4. Place head TLT sphere on phantom ring as shown in Figure 3-2-2-3.
5. Slide the coil all the way forward (away from the magnet) on the base plate and set the locks as shown in Figure 3-2-2-4.
6. Establish an axial landmark on the line on top of the coil. Advance to scan.
7. End the previous exam if this has not already been done.

Figure 3-2-2-1

Slide Coil Back to Access Head Holder

Figure 3-2-2-2

Place Phantom Ring on Head Holder

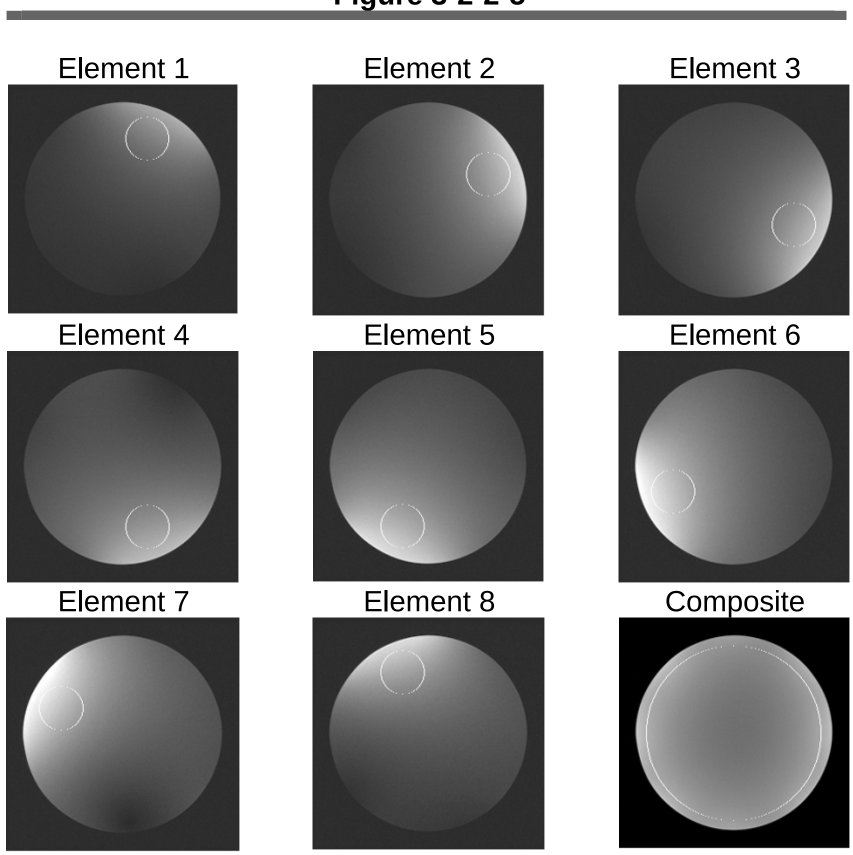
Figure 3-2-2-3

Place Head TLT Sphere on Phantom Ring

Figure 3-2-2-4

Sphere Aligned with Coil

Figure 3-2-2-5



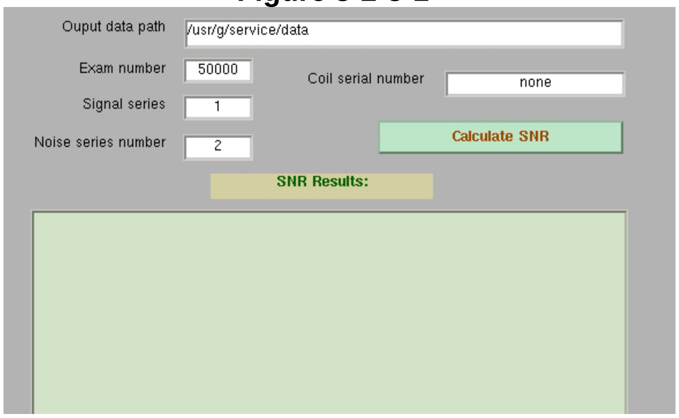
Individual Element ROIs

Continue for 14x Software Versions or Greater	**Continue for 11x and 12x Software Versions**
8. On the Common Service Desktop, select Image Quality, and then the MCQA tool. 9. Make the proper coil selection and press the “Start Scan” button.	8. From the service browser, select Image Quality **->** Coil SNR **->** Click here to start this tool 9. Select the coil type you are testing, enter the serial number and select Start Scan. **Note:** Additional documentation can be accessed by selecting File **->** Help **->** 8HRBRAIN **Note:** The SNR tool requires that the **8HRBservice** service configuration file be installed. The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only. **Proceed to 3-2-5 Acceptance Limits**

3-2-3 Data Analysis (10.X Only)

1. In the command window enter **snr_for_8hrbrain_QA** and press enter. A screen like that shown in Figure 3-2-3-1 should appear.

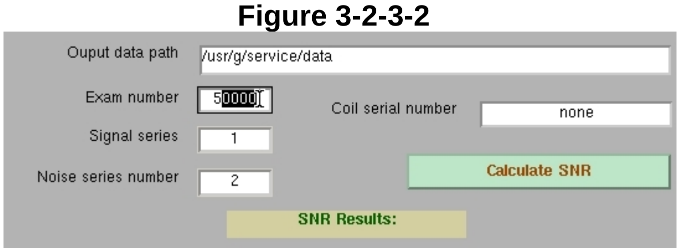
Figure 3-2-3-1



QA Mode SNR Analysis Screen

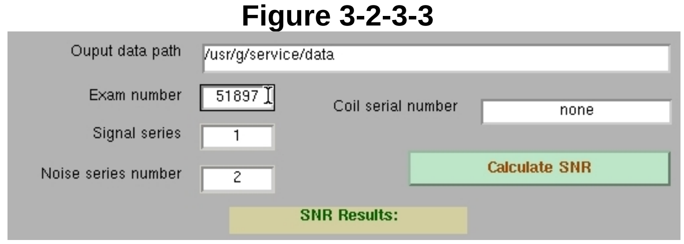
2. In the Exam number window enter the exam number as shown in Figures 3-2-3-2 and 3-2-3-3 (The Delete key may be used instead of Backspace).
3. In the Coil serial number window, enter serial number and bay number as shown in Figures 3-2-3-4 and 3-2-3-5 (Optional).

Figure 3-2-3-2



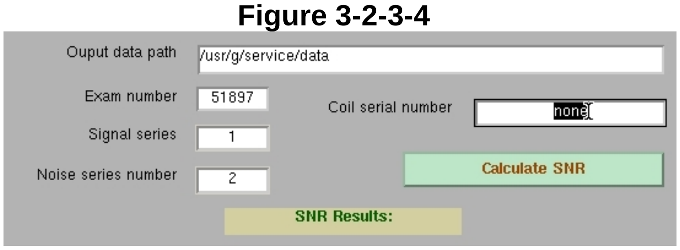
Highlight Digits for Exam Number

Figure 3-2-3-3



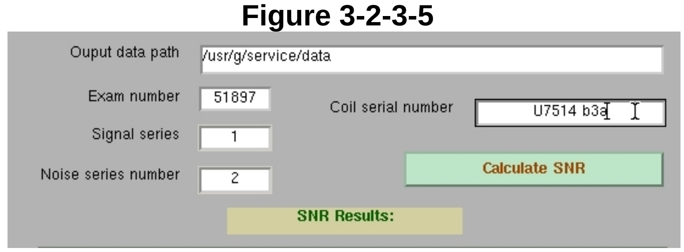
Type In Exam Number

Figure 3-2-3-4



Highlight Coil Serial Number

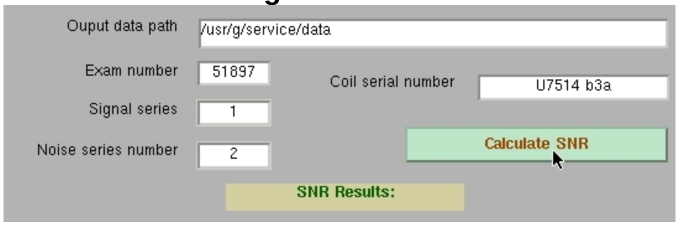
Figure 3-2-3-5



Enter Coil Serial Number and Bay Number

4. Click the Calculate SNR button as shown in Figure 3-2-3-6.

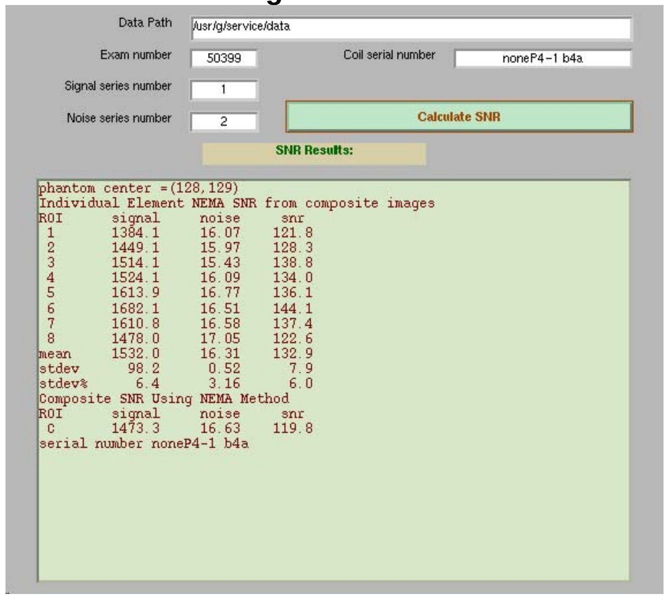
Figure 3-2-3-6



Click Calculate SNR Button

5. Results will appear as shown in Figure 3-2-3-7.
6. Table 3-2-3-1 shows the test results that correspond to each test. For further explanation, refer to Section 3-2-4, Plotting Tool Procedure.

Figure 3-2-3-7



QA Mode SNR Results

EXAMPLE OUTPUT OF TOOL – TABLE 3-2-3-1

```
phantom center =(131,137)
Individual Element NEMA SNR from composite images
ROI       signal     noise     snr
 1        1835.0     19.39    133.8
 2        2054.6     19.61    148.1
 3        2117.0     19.44    154.0
 4        2005.0     19.79    143.3
 5        1994.5     19.49    144.7
 6        2080.0     19.70    149.3
 7        2029.9     20.24    141.8
 8        1848.8     20.19    129.5
mean      1995.6     19.73    143.1
stdev      102.8      0.33      8.1
stdev%       5.2      1.66      5.7
Composite SNR Using NEMA Method
ROI       signal     noise     snr
 C        1920.1     19.99    129.8
serial number U7514 bay3a
```

Using ROIs shown in Figure 3-2-2-5 signal is measured on each of the two composite images and averaged. The same ROI for each element is used to measure noise on the one filtered difference image

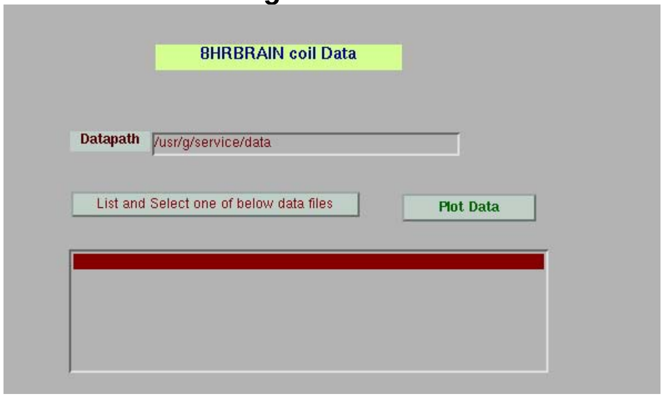
Composite image results

Note: The results are also stored in a file in the /usr/g/service/data directory. The filename will be similar to 8hrbrain_51897_c7ga0522.qa where 51897 is the exam number.

3-2-4 Plotting Tool Procedure (10.X Only)

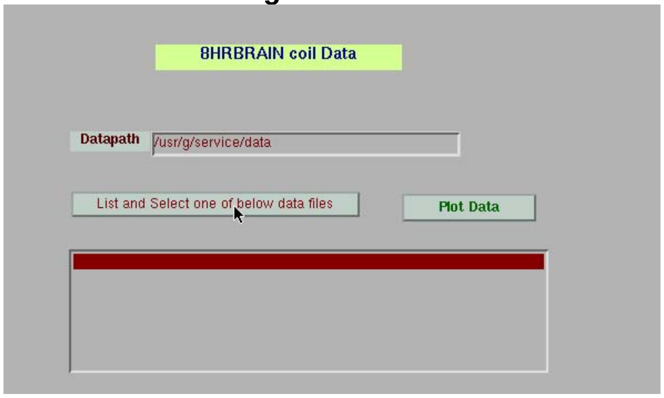
1. Open a command window on the tools desktop.
2. Navigate to the /usr/g/service/cclass directory.
3. Type the command **plot_8hrbrain_data** and press the enter key.
4. The screen shown in Figure 3-2-4-1 will appear.
5. The path to data files, /usr/g/service/data, should be selected by default. If it is not, highlight the text in the type in the correct path.
6. Click on the “List and Select one of below data files” button as shown in Figure 3-2-4-2.
7. Click on the name of the file to be plotted as shown in Figure 3-2-4-3.
8. Click on the Plot Data button as shown in Figure 3-2-4-4 to process the file.
9. Example results for a nominal coil are shown in Figure 3-2-4-5. Coil with poor SNR in Figure 3-2-4-6. Coil with shorted blocking diode in Figure 3-2-4-7. It is interesting to note that the coil whose results are shown in Figure 3-2-4-7 marginally passes the composite image SNR limits but that element 3 performance in individual element results is severely out of normal range.

Figure 3-2-4-1



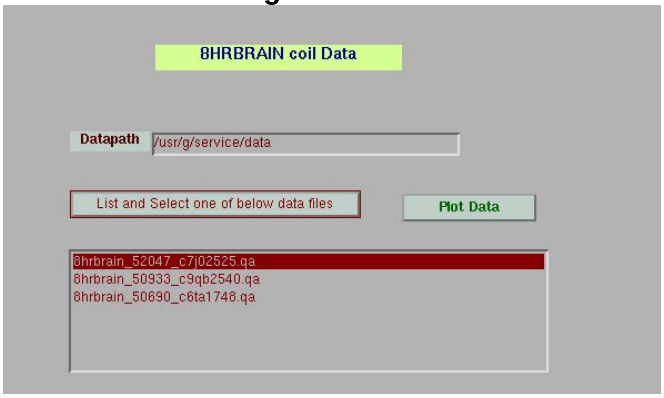
Enter Datapath

Figure 3-2-4-2



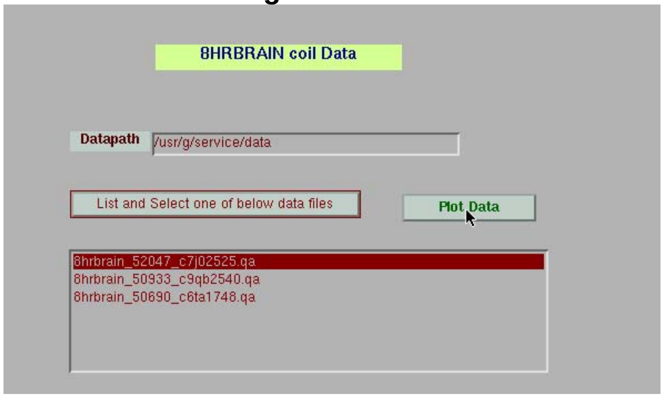
List Files

Figure 3-2-4-3



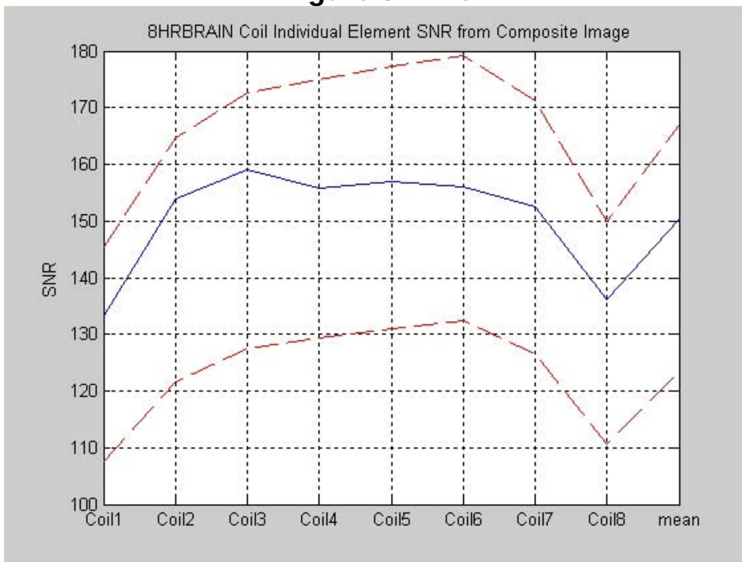
Select File

Figure 3-2-4-4



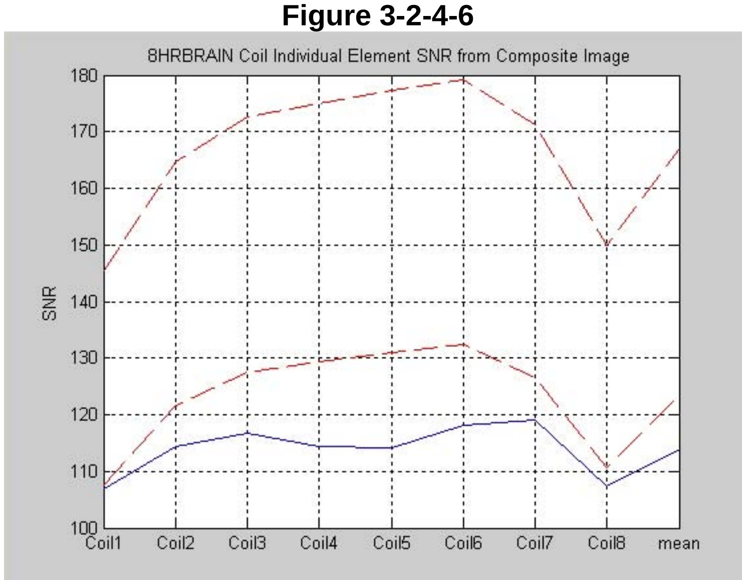
Process File

Figure 3-2-4-5



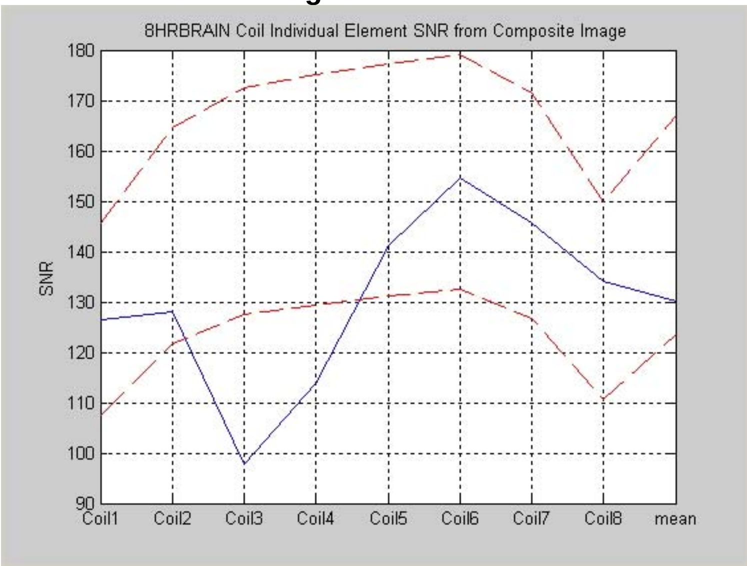
Nominal Individual Element Results

Figure 3-2-4-6



Low SNR Individual Element Results

Figure 3-2-4-7



Shorted Blocking Diode Individual Element Results

3-2-5 Acceptance Limits (10.X and 11.X)

1. Record the coil serial number and the associated filename in Table 3-2-5-1 for future reference.

   Note: The analysis tool can be used to process multiple runs by entering a new exam number and coil serial number and pressing the Calculate SNR button again. It is not necessary to exit the analysis tool after each run.

2. Using Table 3-2-3-1 as a reference, compare the test results to the values in Tables 3-2-5-2 and 3-2-5-3. For additional clarification, refer to Section 3-2-4, Plotting Tool Procedure.

COIL SERIAL NUMBER AND FILENAME – TABLE 3-2-5-1

Date	Coil Serial Number	Filename

COMPOSITE SNR ACCEPTANCE LIMITS – TABLE 3-2-5-2

Nominal	Lower Limit	Upper Limit
131.3	118.2	144.5

INDIVIDUAL ELEMENT SNR LIMITS – TABLE 3-2-5-3

	element	nominal	lower limit	upper limit
individual elements from composite image	1	126.6	107.6	145.6
	2	143.2	121.7	164.6
	3	150.1	127.6	172.6
	4	152.2	129.4	175.1
	5	154.2	131.0	177.3
	6	155.8	132.5	179.2
	7	149.0	126.6	171.3
	8	130.3	110.8	149.9
	mean	145.2	123.4	166.9

3-2-6 Test Explanation and Results Interpretation

The 8 channel hi res brain coil is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses: Separate signal and noise images are evaluated for individual element signal, noise and SNR, much like other automated surface coil tests such as those used for both types of 1.5T CTL coils. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference images. Both types use the same individual element regions of interest (ROIs) which are shown in Figure 3-2-2-5. The Composite SNR using NEMA method also uses the ROIs shown in Figure 3-2-2-5. Only results from composite images are displayed to the user.

- Composite SNR acceptance limits are shown in Table 3-2-3-3.
- Example output of the tool is shown in Table 3-2-3-1.

An automated plotting tool is available to evaluate a particular result file against the individual element acceptance limits shown in Table 3-2-3-4. See Section 3-2-4, Plotting Tool Procedure, for instructions. Example plots from good and defective coils are shown in that section.

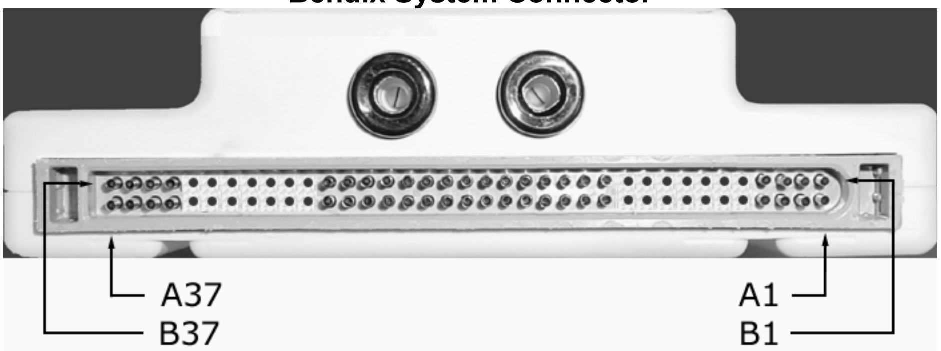
3-3 External Cable Check

Each MCP signal connection has a DC blocking capacitor in the cable connector. Therefore, continuity of the RF signal conductors cannot be determined in the usual way, by measuring from pin to pin using a multimeter. The external cable check must be performed in two parts, section 3-3-1 for the RF signal conductors and 3-3-2 for the PIN diode control conductors.

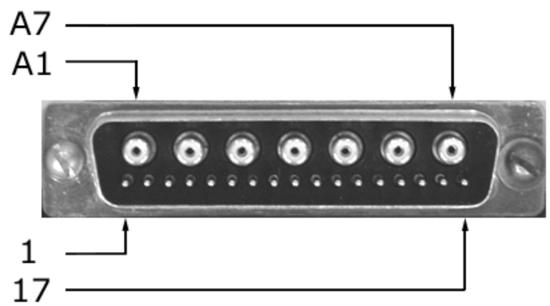
Use Figure 3-3 for pin identification while performing the procedures in sections 3-3 and 3-4.

System and Coil Connectors Pin Identification Figure 3-3

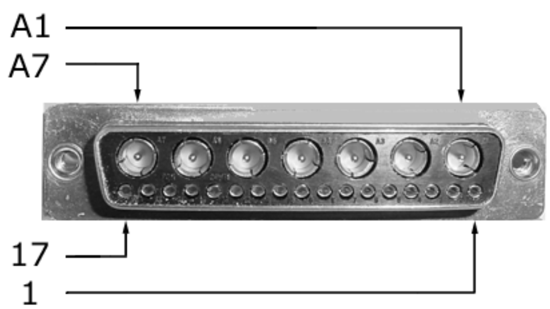
Bendix System Connector



SubD Coil Cable Connector



SubD Coil Connector



3-3-1 RF Signal Channels

The following procedure will rule out a short to GND for each of the 16 signal pins detailed in Table 3-3-1. This check does not, however, determine continuity for each of the MCP signal connections.

Check the coil cable using the following procedure. Referring to Section 5-2, External Cable Replacement, disconnect the SubD coil cable connector. Using a digital multimeter, measure each of the 16 signal pins designated in Table 3-3-1 with respect to GND. Verify that the signal pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-1

Signal Channel	SubD Coil Cable Connector	GND
MCP1	A7	1,3 and shield
MCP2	A6	
MCP3	A5	
MCP4	A4	
MCP5	A3	
MCP6	A2	
MCP7	2	
MCP8	A1	
Signal Channel	Bendix System Connector	GND
MCP1	B1	A12, A14, A16, A18,A20, A22, A24 and A26
MCP2	A2	
MCP3	B3	
MCP4	A4	
MCP5	B34	
MCP6	A35	
MCP7	B36	
MCP8	A37	

If one or more of the pins are shorted to GND, replace the coil cable.

3-3-2 PIN Diode Control Channels

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control channel in Table 3-3-2. Also verify that the control pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-2

Control Channel	SubD Coil Cable Connector	Bendix System Connector
MC 1	17	B12
MC 2	16	B14
MC 3	15	B16
MC 4	14	B18
MC 5	4	B20
MC 6	5	B22
MC 7	6	B24
MC 8	7	B26
GND	1,3 and shield	A12, A14, A16, A18, A20, A22, A24 and A26

If one or more of the readings are > 3 ohms, replace the coil cable. If all readings are < 3 ohms, proceed to the **Diode Check.**

3-4 PIN Diodes Check

With the coil cable connected to the coil, use a multimeter to measure the diode drop across the coil connector pins detailed in Table 3-4. Use Figure 3-3 for pin identification while performing this procedure.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Bendix System Connector		Reading
Positive Lead	**Negative Lead**	
B12	GND [A12]	1.7 ± 0.2
GND [A12]	B12	Open
B14	GND [A14]	1.7 ± 0.2
GND [A14]	B14	Open
B16	GND [A16]	1.7 ± 0.2
GND [A16]	B16	Open
B18	GND [A18]	1.7 ± 0.2
GND [A18]	B18	Open
B20	GND [A20]	1.7 ± 0.2
GND [A20]	B20	Open
B22	GND [A22]	1.7 ± 0.2
GND [A22]	B22	Open
B24	GND [A24]	1.7 ± 0.2
GND [A24]	B24	Open
B26	GND [A26]	1.7 ± 0.2
GND [A26]	B26	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo *3-2 Coil Imaging Performance Verification*. If the problem still exist, replace the 1.5T 8 Channel High Resolution Brain Array.

3-5 Mechanical Hardware Check

Not Applicable

3-6 Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware: *The prescribed RF coil and detected RF coil do not match.*

Trouble-shooting instructions are available online by accessing the service browser: **Diagnostics -> Peripherals -> SRI Functional Tests -> Coil ID** (click directly on the text).

Start the diagnostic by selecting **SRI Coil ID** and **Run**. The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the trouble-shooting instructions if the correct ID is not displayed. When the trouble-shooting is complete, and if the correct ID is still not displayed, replace the cable.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The GE SIGNA 1.5T 8 Channel High Resolution Brain Array, Baseplate and Head Pads may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

None

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

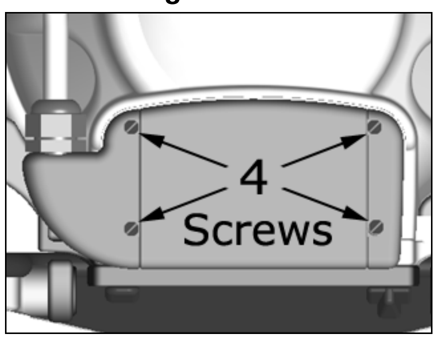
5-1 Disassembly of Coil

Not Applicable

5-2 External Cable Replacement

The 4 coil connector mounting screws are located at the superior end of the coil. Remove the 4 coil connector mounting screws as detailed in Figure 5-2-1. Remove the coil connector by pulling away from the coil in the superior direction.

Figure 5-2-1



Remove 4 Screws

Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

5-3 Mechanical Hardware Replacement

Not Applicable

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	INVIVO #
1.5T 8 Ch HiRes Brain Array / Baseplate Assm.	2317112-2	101844
Cable	2317112-8	101853
Phantom Positioner	2317112-3	102558
1.5T 8Ch HiRes Brain / Latch and Coil Adapter Assy.	2317112-12	104537

Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	INVIVO #
Thick Head Pad	2317112-9	102985
Thin Head Pad	2317112-10	102984
Spacer Pad	2317112-7	101622
Wedge Pad	2317115-8	101407

SECTION 7 – APPENDIX

7-1 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation.

7-2 Coil Configuration

Check the system coil configuration file for the 8HRBRAIN coil configuration. If 8HRBRAIN has not been installed, select 8HRBRAIN from the list of available coils, and using the config file manager, install the following coil names: 8HRBrain and 8HRBService.

Revision History

Rev	Date	Author	Primary Reason for Change
A	11-20-01	L. Hyler	Initial Release
B	04-15-02	L. Hyler	Completed External Cable Check & PIN Diode Check
C	06-12-02	L. Hyler	Complete Functional Checks, Section 3
D	07-29-02	L. Hyler	Make Corrections Outlined in GE Formal Review Log 07-26-02
			Add Automated QA Section
E	08-13-02	L. Hyler	Increased Length of Table 3-2-3-2
F	08-20-02	L. Hyler	Corrected Minor Typos
01	08-23-02	L. Hyler	Initial Issue
01A	12-05-02	L. Hyler	Update QA Procedure per GE
01B	01-09-03	L. Hyler	Replace Figures 3-2-4-5, 3-2-4-6, 3-2-4-7
01C	01-09-03	L. Hyler	Add text back to Section 3-2-4 item 9
02	01-15-03	L. Hyler	Release changes – 01A, 01B, 01C
03	05-19-03	L. Hyler	Replace Head Pad Part Numbers with Thin Head Pad and Thick Head Pad Part Numbers
04	06-18-03	L. Hyler	New Baseplate Illustrations
			Section 3-2-2 correct path to **/usr/local/bin/CoilsTools**
05	08-11-03	L. Hyler	Changed Table 3-4 spec. from 2.0 ± 0.2 to 1.7 ± 0.2
06	11-04-03	L. Hyler	Redesigned section 3-2-2 for both version 10.X and 11.X s/w Add Coil ID Information to Sections 3-6, 5-2, and 6-1
07	04-02-04	L. Hyler	Added Latch and Coil Adapter Assy., Section 6-1
08	07-06-05	D. Visscher	Rebrand MRI Devices with Invivo (DCN 1374)
9	6/19/08	V. Thacker	Customer Change Request CCN0021 (DCN3994)
10	6/24/09	R.M. Scott	Add Coil Configuration (DCN 5847)